

SC-SCOPE endpoint: the joint incompatible-pairing argument is exhausted (strengthened honest negative)

TECT verification-first repository · claim B5-BEYOND-LAYER-BOUND

first issued 2026-06-07 · this version issued 2026-06-07 · v1.0

1. Purpose and scope

The joint-assessment note (scscope-endpoint-joint-assessment v1.0) found the additive endpoint joint at $\times 0.757 < 1$ and registered the joint incompatible-pairing argument as the route to recover the $\times 1.32$ deficit. This note carries out that route and reports the outcome: it does NOT close the endpoint. Scope: the $I = 2 \times 10^{-3}$ endpoint; the pairing is evaluated in its most favourable form, so the conclusion is robust.

2. The pairing argument and why it is the best case

The two surviving third-order channels peak at different transfers: the sunset per-transfer cost rides $J(t)$ (largest at small t), the quartic-difference inflation $R(t) \propto \widehat{G^4}(t)/J(t)$ peaks at $t = 2q_0$ (scscope_ghat4_pertransfer). Modelling

$$R_s(t) = R_{s,\max} \frac{J(t)}{J(0)}, \quad R_q(t) = R_{q,\max} \frac{\Phi(t)}{\Phi_{\max}}, \quad R_{s,\max} = 1.42, \quad R_{q,\max} = 1.02,$$

with maximally-separated peaks, the joint inflation is bounded by $\max_t [R_s(t) + R_q(t)]$ rather than the additive $R_{s,\max} + R_{q,\max}$. Because the peaks are maximally separated, this model gives the SMALLEST possible joint max: any true per-transfer structure has peaks no further apart, so the true joint inflation is $\geq$ this. Hence a non-closure here is unconditional.

3. Result: the endpoint still does not close

Over the realized chords,

$$\max_t [R_s(t) + R_q(t)] = 1.872 < R_{s,\max} + R_{q,\max} = 2.435$$

(the pairing genuinely helps), but

$$\text{paired endpoint} = \frac{\rho}{1 + \max_t [R_s + R_q]} = \frac{2.6}{2.872} = \times 0.905 < 1.$$

The maximum sits at the smallest transfer $t = t_{\min}$, where the sunset is near its peak. The root cause is explicit: the sunset ALONE gives $\rho/(1 + R_{s,\max}) = \times 1.076$ – the third-cumulant sunset already nearly saturates the $\times 2.6$ second-order floor at the thinnest endpoint, so adding even the small- t tail of the quartic-difference ($R_q(t_{\min}) = 0.63$) pushes the joint below unity. This is a structural largeness of the third cumulant at $I = 2 \times 10^{-3}$, not a loose-pairing artefact.

4. Consequence: per-transfer refinements are exhausted

The per-transfer programme that resolved M-ENDPOINT (sunset coupling) and GHAT4-PERTRANSFER (quartic form factor) and now the joint pairing has been carried to completion; it improves the endpoint from the sup-grade $\times 1.0$ inventory to $\times 0.905$ but does not reach unity. The SC-SCOPE all-orders ENDPOINT therefore does not close by per-transfer/pairing means. The remaining routes are genuinely different:

- a LARGER second-order STEP-5B endpoint floor (currently $\rho = 2.6$ at $I = 2 \times 10^{-3}$): if the de-thinned floor were $\gtrsim \times 3.9$ the paired joint would close ($3.9/2.872 = \times 1.36$); this is a B5 STEP-5B sharpening, not a third-order task;
- or accepting that the all-orders lift holds for $I \leq 10^{-3}$ (where the floor is $\times 8.8$, paired joint $\times 3.1$) and the selection at the $I = 2 \times 10^{-3}$ endpoint is at second-cumulant scope.

5. Impact

SC-SCOPE stays OPEN at the endpoint; B1 T6 is UNAFFECTED (SC-SCOPE is a named hypothesis – its open status is the tier’s explicit condition). The all-orders lift is now established as FEASIBLE for $I \leq 10^{-3}$ and OBSTRUCTED at the $I = 2 \times 10^{-3}$ endpoint by a genuinely large third cumulant, with the two non-per-transfer routes named. M-ENDPOINT, GHAT4-PERTRANSFER and the pairing analysis are durable; the negative is a sharp map of the obstruction, not a dead end.

6. Devil’s-advocate / self-adversarial review (CLAUDE.md 6.3, 6 code-discipline)

- (α) “*The sunset model $R_s(t) \propto J(t)$ is unproved.*” VALID: it is a model. But it is the MOST FAVOURABLE (maximally separated from the quartic peak); a less favourable structure only worsens the joint. The non-closure is therefore robust to the model.
- (β) “ *$R_{s,\max}, R_{q,\max}$ inherit the cited estimates.*” VALID and bounded: even if $R_{q,\max} \rightarrow 0$ (quartic-difference fully benign), the sunset alone gives $\times 1.076$ – the endpoint margin is intrinsically thin, so the verdict (no comfortable closure) does not hinge on the quartic estimate.
- (γ) “*Does this refute the selection at the endpoint?*” DISMISSED: it leaves the ALL-ORDERS lift open at the endpoint; the second-cumulant selection (the B1 T6 claim) is intact by hypothesis.

Quantitative sanity check (mandatory): magnitude – paired $\times 0.905$ lies between additive $\times 0.757$ and unity, as a partial recovery must; limit-case – $R_q \rightarrow 0$ gives sunset-only $\times 1.076$ (consistent with the root-cause claim); sign-direction – all channels are costs (reduce the margin); monotonicity – the combined $R_s + R_q$ decreases from $t_{\min}$ to $2q_0$, locating the max at $t_{\min}$ as the table shows.

7. Result footer

Result ID:	B5-BEYOND-LAYER-BOUND / SC-SCOPE joint pairing (U16)
Precise statement:	Joint incompatible-pairing (most-favourable model): $\max_t [R_s(t) + R_q(t)] = 1.872 < \text{sum of maxima } 2.435$, but paired endpoint = $\rho / (1 + 1.872) = 2.6 / 2.872 = \times 0.905 < 1$. Sunset alone $\times 1.076$. The SC-SCOPE all-orders endpoint

	does NOT close even under the best pairing; per-transfer refinements are exhausted.
Scope:	I=2e-3 endpoint; favourable-pairing model -> lower bound on the true joint inflation (non-closure unconditional).
Dependencies:	scscope_ghat4_pertransfer (Phi(t), R_q), scscope_joint_endpoint (R_s_max, R_q_max, rho); cited estimates.
Evidence grade:	EXECUTED (scscope_joint_pairing.py 4/4); strengthened honest negative.
Reproduction command:	python codes/vacuum/scscope_joint_pairing.py
Expected output:	joint max 1.872; paired endpoint x0.905; sunset-only x1.076.
Falsification gate:	A STEP-5B endpoint floor $\rho > \sim 3.9$ at I=2e-3 (would close the paired joint) or a genuinely tighter third-cumulant treatment; either CLOSES the endpoint and is the registered route.
Tier before / after:	No tier/gate flip. SC-SCOPE: OPEN at the endpoint (per-transfer/pairing exhausted; x0.905). FEASIBLE for I<=1e-3. B1 T6 unchanged.
No-overclaim statement:	Strengthened honest negative: the endpoint does not close even under the most favourable pairing. The selection is unaffected (second-cumulant by hypothesis). Models stated; conclusion robust to them.
Next required action:	Either a sharper STEP-5B endpoint floor (B5; $\rho > \sim 3.9$ at I=2e-3) or formal acceptance of second-cumulant scope at the I=2e-3 endpoint -- an operator decision.